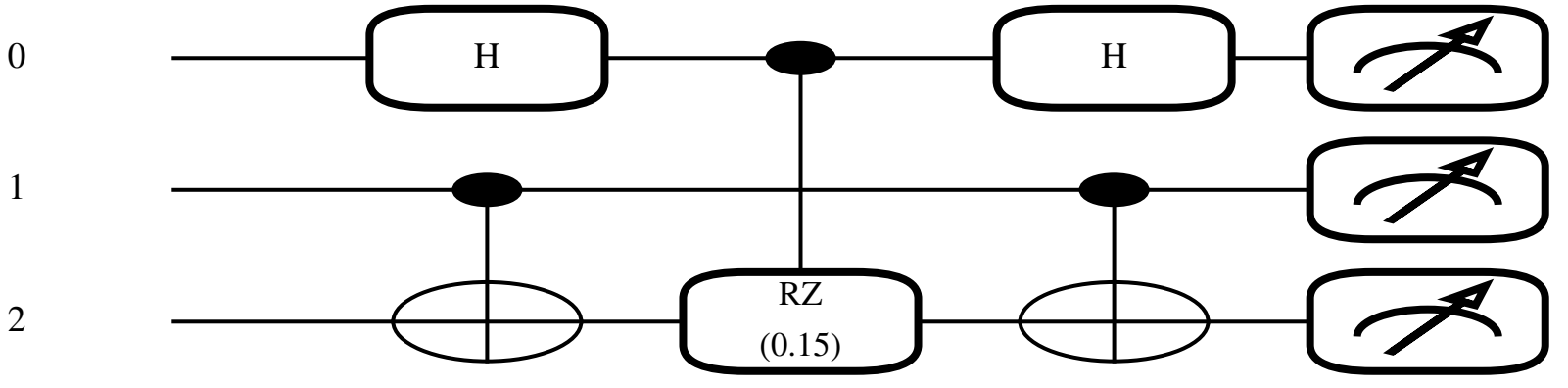
Single Walsh factor circuit: $\hat{U}_3 = e^{-i\varepsilon a_3^\star Z_1 Z_2}$



$\theta_1 = 2\varepsilon a_3^\star = 0.148$

Implemented by parity compute, controlled-$R_Z(\theta_1)$, and uncompute.